Detecting Misaligned Product Images in E-commerce Reviews Using Fine-Tuned CLIP

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Abstract

Large-scale pre-trained models like CLIP have shown promising capabilities in bridging vision and language tasks. However, their performance on noisy, user-generated data in real-world scenarios remains underexplored. This study investigates fine-tuning CLIP with two different loss functions—contrastive and cross-entropy—to evaluate its potential for detecting semantic misalignment in image-text pairs. Using a dataset curated from Trendyol's product links, consisting of both official product images and user-uploaded photos, the models were evaluated under noisy and misaligned conditions. The cross-entropy fine-tuned model achieved an accuracy of 85%, while the contrastive model reached 79%, providing preliminary insights into the adaptability of multi-modal models for real-world, imperfect data. The findings highlight both the potential and the limitations of such approaches, underscoring the need for further refinement and exploration.

1 Introduction

Pre-trained multi-modal models such as CLIP [9] have demonstrated remarkable performance in bridging vision and language tasks. However, their robustness in real-world scenarios, where data is often noisy and user-generated, remains underexplored [6]. Real-world datasets often include challenges such as low-quality images, incomplete text descriptions, or misaligned semantics between modalities. These limitations are particularly evident in user-generated content, where inconsistencies and noise are prevalent, necessitating the adaptation of models like CLIP to handle such imperfections effectively.

In this study, I focus on fine-tuning the CLIP model to classify image pairs as either aligned or misaligned, with a specific emphasis on detecting discrepancies between a product's official image and user-uploaded images in e-commerce contexts. Such a capability could assist in identifying irrelevant or misleading visual content uploaded by users. Due to data privacy regulations, such as KVKK (Turkey's equivalent of GDPR), user comments and images were not directly scraped in an automated fashion. Instead, I collected the official product images programmatically using a web-scraping script, while the user-uploaded review images were gathered manually.

Thresholds for classification were determined empirically during experimentation. The choice of a threshold value (0.4) reflects a balance between precision and recall, achieving a classification accuracy of 75%. Although this value indicates potential, it is acknowledged that the limited size and variability of the dataset, as well as computational constraints, present opportunities for further refinement.

The contributions of this study can be summarized as follows:

- A lightweight fine-tuning approach for CLIP to classify noisy image pairs, focusing on practical use cases such as identifying misaligned user-uploaded images in e-commerce.
- An evaluation of CLIP's performance with curated datasets, reflecting the complexities of real-world data, and achieving 75% accuracy under constrained computational resources.
- A discussion of the challenges associated with multi-modal alignment under noisy conditions, including ethical considerations and practical implications of thresholds for classification tasks.

2 Related Work

Pre-trained multi-modal models such as CLIP [9] have achieved state-of-the-art performance in vision-language tasks, including image classification, retrieval, and captioning. These models rely on large-scale training datasets to learn robust joint embeddings for vision and language. However, their robustness in noisy, user-generated scenarios, where data may include semantic misalignments or degraded quality, has not been thoroughly explored.

2.1 Contrastive Learning and Multi-Modal Alignment

Contrastive learning methods, such as SimCLR [1], MoCo [4], and ALIGN [6], have been foundational in improving visual and multi-modal representation learning. These approaches rely on large-scale, high-quality datasets for robust alignment but are not often evaluated under real-world noisy conditions. Recent models such as RobustCLIP [8] incorporate synthetic noise to improve robustness, though their application in naturally noisy scenarios remains limited.

CLIP and ALIGN are widely adopted for vision-language alignment tasks, yet they focus on global matching problems like retrieval rather than fine-grained classification. Additionally, these models assume high-quality data during pretraining, which may not generalize well to real-world, user-generated content. Fine-tuning such models to handle noise and misalignment represents an under-explored challenge.

2.2 Applications in E-Commerce and Noisy Data Domains

In the e-commerce domain, multi-modal learning has been utilized to improve tasks like product search and recommendation. Hu et al. [5] address noisy text descriptions in e-commerce by introducing an online text-cleaning approach, while CommerceMM [7] proposes a large-scale framework for product retrieval and recommendation. Similarly, Dong et al. [2] focus on predicting customer satisfaction using multi-modal data in e-commerce, highlighting the role of vision and language integration.

While these studies emphasize multi-modal learning for structured e-commerce data, they do not address unstructured or user-generated content, such as mismatched images uploaded by users. A recent trend in e-commerce platforms allows users to share photos of the products they purchase, enhancing customer engagement and transparency. However, this feature introduces challenges, as users may upload irrelevant images—either unintentionally, for incentives like coupons, or maliciously in cases involving fraudulent activities. This gap highlights the need for models capable of effectively detecting and handling such noisy and misaligned data to maintain the trustworthiness and integrity of these platforms.

Beyond e-commerce, prior works on noisy data in multi-modal tasks have targeted specific domains like social media [10] and surveillance [11]. These studies primarily aim to address domain-specific challenges and often rely on synthetic noise for robustness evaluation. The generalization of these approaches to diverse, user-generated data remains a significant challenge.

2.3 Evaluation of User-Generated Data

This study investigates the adaptation of CLIP for noisy image-text alignment using curated datasets that approximate real-world user scenarios. While the data used in this work includes structured elements such as product images and captions, it also incorporates aspects of user-generated content that introduce variability and noise. Unlike prior studies that predominantly rely on well-curated

datasets or synthetic noise, this work leverages manually scraped data to mimic the types of challenges encountered in practical applications. The goal of this exploration is not to provide a comprehensive solution but to examine the potential of pre-trained multi-modal models like CLIP to handle moderately noisy data and highlight areas where further refinement is needed.

3 Methodology

3.1 Dataset Construction

The dataset for this study was constructed by leveraging the [Trendyol Product Dataset] available on Kaggle . This dataset originally consisted of over 30,000 product links, which were used to scrape images and associated text descriptions from Trendyol's website. Product images were retrieved programmatically by parsing the HTML source code of the product pages, while user-uploaded images linked to these products were manually collected due to their less-structured format. After filtering invalid links and unusable pairs, the final dataset consisted of over 22,000 image-text pairs.

The web scraping and data preparation were conducted on an M4 Mac Mini, taking approximately 6 hours to complete.

3.2 Loss Functions and Model Fine-Tuning

To adapt the CLIP model for the task, two separate models were fine-tuned, each leveraging a different loss function. The use of separate models was essential to evaluate the impact of each loss function independently and to compare their effectiveness in addressing the challenges:

- **Contrastive Loss**: This loss function leverages the inherent alignment capabilities of the CLIP model's embedding space. Cosine similarity between image and text embeddings is computed, where aligned pairs are encouraged to exhibit higher similarity, while misaligned pairs are penalized using a margin-based formulation. This approach preserves the model's ability to represent semantic alignment while incorporating robustness against noise.
- **Cross-Entropy Loss**: A separate model was trained as a binary classifier to predict whether a given image-text pair is aligned or misaligned. By framing the task as a classification problem, this approach enhances the model's ability to explicitly differentiate between aligned and misaligned pairs, making it suitable for tasks where direct pair classification is required.

By training distinct models with these two loss functions, it was possible to explore their strengths and limitations in noisy environments, offering valuable insights into their applicability for real-world, imperfect data.

3.3 Training and Optimization

The training configurations were as follows:

- **Contrastive Loss Model**: Fine-tuned using the AdamW optimizer with a learning rate of $1e^{-6}$ and a batch size of 64. Training lasted for 3 epochs (approximately 6 hours). The low learning rate ensured minimal disruption to CLIP's pre-trained embedding space, mitigating the risk of catastrophic forgetting.
- **Cross-Entropy Loss Model**: Fine-tuned with the AdamW optimizer, a learning rate of $1e^{-5}$, and a batch size of 64. This setup was chosen to allow faster convergence for binary classification tasks. Training took approximately 8 hours for 3 epochs.

AdamW was chosen due to its decoupled weight decay mechanism, which prevents overfitting by effectively regularizing large model weights. This makes it particularly suitable for fine-tuning transformer-based models like CLIP. Alternative optimizers such as SGD and RMSprop were considered but found less appropriate: SGD often requires extensive tuning and converges slower, while RMSprop is not ideal for the gradient dynamics of large-scale transformer architectures.

Table 1: Performance Comparison

Model	Accuracy	F1-Score	Precision (A / M)	Recall (A / M)
Original CLIP	0.68	0.65	0.70 / 0.65	0.60 / 0.70
Fine-Tuned CLIP (Contrastive Loss)	0.79	0.79	0.76 / 0.82	0.84 / 0.74
Fine-Tuned CLIP (Cross-Entropy Loss)	0.85	0.84	0.80 / 0.91	0.92 / 0.78

Alternative Approaches Although this study focuses on separate fine-tuning tasks, other methods could enhance performance:

- **Dynamic Learning Rates**: Schedulers like cosine annealing could better guide optimization but were not employed due to the limited training epochs (3 epochs).
- **Unified Multi-Task Learning**: A combined framework for optimizing both contrastive and cross-entropy losses might improve efficiency but was beyond the scope of this evaluation.

Checkpoints were saved every 100 steps, enabling progress tracking and recovery. At the end of each epoch, loss and accuracy metrics were calculated to monitor performance trends and ensure alignment with objectives. Due to time and processing limitations, training was limited to 3 epochs, though extended training on higher-performance systems may yield improved results.

4 Experiments

4.1 Experimental Setup

The evaluation of the fine-tuned CLIP model was conducted using a dataset derived from the Trendyol Product Dataset. The dataset contained over 20,000 valid image-text pairs after removing invalid links. Product images were programmatically scraped, while user-uploaded images associated with the products were manually curated due to their unstructured format.

Baseline comparisons were performed using the following models:

- **Original CLIP**: The pre-trained CLIP model was evaluated directly without any fine-tuning.
- **Fine-Tuned CLIP with Cross-Entropy Loss**: A classifier fine-tuned with a cross-entropy loss to directly classify image-text pairs as aligned or misaligned.
- **Fine-Tuned CLIP with Contrastive Loss**: A separate model fine-tuned with contrastive loss to retain embedding alignment capabilities.

4.2 Evaluation Metrics

The models were evaluated on the following metrics:

- **Accuracy**: The proportion of correctly classified pairs.
- **F1-Score**: A balanced measure combining precision and recall.
- **Precision and Recall**: Assess the ability to correctly identify aligned and misaligned pairs.

5 Results and Analysis

The performance of the evaluated models is summarized in Table 1. Metrics were computed on a test set of 9,215 image-text pairs, evenly split between aligned and misaligned categories.

Observations

- **Fine-Tuned Models Outperform Original CLIP**: Both fine-tuned models demonstrated significant improvements over the original CLIP, emphasizing the importance of task-specific adaptation for noisy datasets.

- **Cross-Entropy Loss Model Achieves Best Overall Performance**: The cross-entropy model achieved the highest accuracy (85%) and F1-score (0.84), making it more effective for binary classification tasks under real-world noisy conditions.
- **Challenges with User-Generated Noise**: Both fine-tuned models effectively handled real-world noise, such as unrelated product images uploaded by users. However, there remains room for improvement, particularly in misaligned cases.

6 Discussion

Training Configuration and Constraints Both models were trained for 3 epochs on an M4 Mac Mini, with approximately 8 hours per model. This limitation was primarily due to computational constraints, and longer training durations could potentially enhance performance. Furthermore, the small size of the curated dataset may have limited the generalizability of the results, suggesting a need for larger datasets in future work.

Comparison to Baselines The original CLIP model achieved only 68% accuracy, struggling to generalize on noisy, user-generated data. By contrast:

- The cross-entropy loss model excelled in binary classification tasks, leveraging its discriminative training objective to achieve 85% accuracy.
- The contrastive loss model, while slightly less accurate, maintained a strong balance of precision and recall, making it suitable for tasks requiring embedding alignment and similarity computations.

Potential Improvements Several improvements could be explored to build on the current results:

- **Extended Training**: Increasing the number of epochs and utilizing higher computational resources could allow the models to better converge on the noisy data.
- **Augmented Datasets**: Incorporating adversarially generated noise and more diverse user-uploaded data could improve robustness.

Real-World Implications These findings suggest practical applications in scenarios such as e-commerce platforms, where detecting misaligned user-uploaded images is critical to ensuring data integrity and user trust. The cross-entropy model's superior classification performance and the contrastive model's embedding robustness offer complementary strengths that could be tailored to specific downstream tasks.

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